MIL-B-17095G
7 April 1987
SUPERSEDING
MIL-B-17095F
12 September 1983

MILITARY SPECIFICATION

BOILERS, STEAM, WATERTUBE, PACKAGED TYPE (OUTPUT 5,000,000 TO 125,000,000 BTU/HR, OPERATING PRESSURE 100 TO 900 PSIG)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers shop assembled (packaged type), multiple drum, natural circulation watertube, steam boilers equipped for oil, gas, or combination oil-and-gas firing. The boilers covered by this specification are intended for generation of steam to be used in processing operations, space heating, or power generating systems on land installations.

1.2 Classification. The boilers shall be of the following types:

1.2.1 Firing system types. The boilers shall be of the following firing system types with specific type of fuel to be burned, as specified (see 6.2.1).

Type LO	- Light-oil fired (Grade 2, diesel, or JP-5, as specified).
Type HO	- Heavy-oil fired (Grade 4, Grade 5 (light), Grade 5 (heavy), Grade 6, or Navy special, as specified).
Type G	- Gas-fired (natural, manufactured, mixed, or liquefied petroleum gas, as specified).
Type GLO	- Combination gas-fired and light-oil fired (type of gas and grade of oil, as specified).
Type GHO	- Combination gas-fired and heavy-oil fired (type of gas and grade of oil, as specified).
Type GLOHO	- Combination gas-fired, light-oil, and heavy-oil fired (type of gas and grades of oils, as specified) (see 6.1.1d).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Control sequences. The boilers shall have the following control sequences, as specified (see 6.2.1).

Control sequence I - Automatic system (recycling).
Control sequence II - Automatic system (non-recycling).
Control sequence III - Supervised manual system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture and Fungus Resistant (For the Treatment of Communications, Electronics, and Associated Electrical Equipment).
MIL-B-3180 - Boilers and Related Equipment, Packaging of.
MIL-R-18115 - Regulators, Boiler-Feed-Water, Automatic.
MIL-V-18406 - Valves: Blowoff, Boiler.
MIL-B-18796 - Burners, Single, Oil, Gas, and Gas-Oil Combination (400,000 Btu/hr and Over Input Capacity).
DoD-P-24555 - Paint, Aluminum, Heat-Resistant (650°C) Low-Emissivity (0.40 or less) (Metric).

STANDARDS

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptability Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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CODE OF FEDERAL REGULATIONS (CFR)

Title 40, Environmental Protection

Part 60 - New Stationary Source Standards, General Provisions for Fossil-fueled Steam Generators/Incinerators Larger than 50 TPD.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are Department of Defense (DOD) adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN BOILER MANUFACTURERS ASSOCIATION (ABMA)

Lexicon Steam Generating Equipment.
Measurement of Sound from Boiler Units, Bottom Supported Shop or Field Erected.
Boiler Water Limits and Steam Purity Recommendations for Watertube Boilers.

(Application for copies should be addressed to the American Boiler Manufacturers Association, 950 N. Glebe Rd., Suite 160, Arlington, VA 22203.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Presure Vessel Code.
Section I - Power Boilers.
Section VIII - Pressure Vessels.

ASME Performance Test Code.
PTC 4.1 - Steam Generating Units.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C64 - Refractories for Incinerators and Boilers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 1 - General Standards for Industrial Controls and Systems.
ICS 2 - Industrial Control Devices, Controllers and Assemblies.
ICS 6 - Enclosures for Industrial Controls and Systems.
MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.
NFPA 85 - Prevention of Furnace Explosions in Fuel Oil- and Natural Gas-Fired Single Burner Boiler-Furnaces.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

STEEL STRUCTURES PAINTING COUNCIL (SPCC)

SP-6 - Commercial Blast Cleaning

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburg, PA 15213.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 296 - Standard for Safety, Oil Burners.
UL 726 - Standard for Oil-Fired Boilers.
UL 795 - Commercial-Industrial Gas-Heating Equipment.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The boilers shall be multiple drum, watertube, natural circulation packaged units. Each boiler shall be equipped with a complete burner and fuel system; a forced draft fan; an automatic combustion control system, flame safeguard controls, firing sequence programmer, safety interlocks, limit controls, and control panel. All units shall be factory-wired and assembled except for such readily installed appurtenances as safety valves, water columns, pressure gages, forced draft fans or blowers, and oil heater piping. These items may be removed prior to shipment to prevent damage, reduce cubage, or meet shipping dimension restrictions. In other respects, the units shall be complete and ready for operation when connected to water, fuel, and electrical supplies. Furnaces for two-drum, D-type boilers may be located either on the right or the left-hand side unless a particular location is preferred or required by the contracting officer (see 6.2.1). Nomenclature of ABMA Lexicon, Steam Generating Equipment and ASME Boiler and Pressure Vessel Code, Section I, Power Boilers, and Section VIII, Pressure Vessels shall apply.

3.2 First article inspection. When specified (see 6.2.1), the contractor shall furnish one boiler for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Standard commercial product. The boiler shall, as a minimum, be in accordance with the requirements of this specification. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the boiler being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, and articles incorporated in the work covered by this specification are to be new.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Design. Boilers shall be designed in accordance with the applicable sections of the ASME Boiler and Pressure Vessel Code and any additional requirements specified herein. Boilers shall be of the conventional bent-tube design for the following conditions, as specified (see 6.2.1).

a. Maximum continuous steam flow in pounds per hour (lb/hr).
b. Maximum allowable steam working pressure in pound-force per square inch gage (psig).

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on the carrier when shipped.

c. Operating pressure (psig) at steam drum or superheater outlet, as applicable.
d. Steam temperature at steam drum or superheater outlet, as applicable in degrees fahrenheit (°F).
e. Feedwater temperature at steam drum (°F).
f. Boiler site elevation (feet).
g. Maximum, minimum anticipated ambient air temperatures (°F).
h. Maximum continuous output in British thermal units per hour (Btu/hr).

3.7 Performance. Unless otherwise specified (see 6.2.1), performance requirements, including furnace heat release rates, shall be based on the following fuel analysis. This analysis is for equipment selection and design purposes and may not meet local environmental emmission requirements (see 6.1.1f).

a. Fuel oil (percent by weight, as fired).

Ultimate analysis:

Hydrogen	10.33
Carbon	87.87
Oxygen	0.25
Nitrogen	0.14
Sulfur	1.16
Ash	0.25
Higher heating value	18,400 Btu/hr

b. Natural gas (percent by volume, as fired).

Ultimate analysis:

Methane	90.0	
Ethane	5.0	
Nitrogen	5.0	
Ash	Nil	
Moisture	0.01	
Higher heating value	1,075	Btu/cubic feet (cu ft)
Higher heating value	21,824	Btu/lb

The gas supply pressure available at the site for burner and gas-train design shall be as specified (see 6.2.1).

3.7.1 Steam quality. The moisture content of the steam leaving the steam drums shall not exceed 0.5 percent for boilers without superheaters when operating under continuous load design conditions with normal water level and stipulated ABMA boiler water concentrations given in table I. The solids content of the steam leaving the steam drum shall not exceed 1 part per million (ppm) for boilers with superheaters when operating under maximum continuous load design conditions with normal water level and stipulated ABMA boiler water concentrations given in table I.

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TABLE I. Boiler water concentrations.

Operating Pressure (psig)	Total Solids (ppm)	Total (P) Alkalinity for $CaCo_3$ (ppm)	Suspended Solids (ppm)
100-300	3,500	700	300
301-450	3,000	600	250
451-600	2,500	500	150
601-750	2,000	400	100
751-900	1,500	300	60

3.7.2 Capacity. The maximum continuous thermal output of the boiler, expressed in Btu/hr at the outlet connection, shall be as specified (see 6.2.1) and shall be based on the conditions as defined in 3.6, and 3.7.1. If the acquisition documentation should specify capacity for boilers in terms of boiler horsepower (hp), the thermal output shall be computed on the basis of 33,475 Btu/hr per boiler hp. The specified capacity shall be attained within the performance limits for heat release specified in 3.7.4.

3.7.3 Efficiency.

3.7.3.1 Combustion efficiency. The carbon dioxide, carbon monoxide, and oxygen content of the flue gas measured at the boiler outlet, when firing the specified fuel (see 3.7 and 6.2.1) shall not exceed the limits specified in table II. When fired with oil specified herein, the boiler-stack emissions shall not exceed 20 percent opacity except for short periods not to exceed 6 minutes per hour. When specified (see 6.2.1), the boilers shall be designed to burn the fuels herein, so that the particulate, opacity and nitrogen oxide emissions shall meet the applicable standards of CFR Title 40, Part 60 (see 6.1.1f and 6.2.1) and shall not exceed the limits determined in accordance with 4.4.3.2.6. Both stable firing and combustion efficiency shall be maintained over the entire firing range required by the burner turndown (see 3.11).

TABLE II. Combustion performance.

Requirements	Natural Gas 1,075 Btu/cu ft			Light Oils No. 2, JP-5, Diesel			Medium Oils No. 4, No. 5 (light), NSFO			Heavy Oils Nos. 5 & 6		
Rated Load (percent)	40	70	100	40	70	100	40	70	100	40	70	100
Carbon dioxide (percent) (minimum)	8.5	9.0	10.0	10.5	11.5	12.5	11.0	12.0	13.0	12.0	12.6	13.6
Oxygen (percent) (maximum)	6.0	5.0	4.0	6.0	5.0	4.0	6.0	5.0	4.0	6.0	5.0	4.0

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TABLE II. Combustion performance - Continued.

Requirements	Natural Gas 1,075 Btu/cu ft			Light Oils No. 2, JP-5, Diesel			Medium Oils No. 4, No. 5 (light), NSFO			Heavy Oils No. 5 & 6		
Rated Load (percent)	40	70	100	40	70	100	40	70	100	40	70	100
Carbon monoxide (percent) (maximum)	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Excess air (percent) (maximum)	40	25	20	40	30	20	40	30	20	40	30	20

3.7.3.2 Boiler unit thermal efficiency. When tested in accordance with 4.4.3.1, the boiler unit thermal efficiency shall not be less than 80 percent for gas-fired and 84 percent for oil-fired boilers (see 6.1.1g).

3.7.4 Furnace heat input rate. When the boiler is operating at maximum output, the heat input rates to the furnace shall not exceed the values specified in 3.7.4.1 and 3.7.4.2.

3.7.4.1 Volume heat input rate. The volume (see 6.1.3.1) heat input rate shall not exceed 64,000 Btu/hr/cu ft of furnace volume for boilers having a total effective radiant heating surface (see 6.1.3.2) equal to 60 percent or less of the total furnace wall area. For each percentage increase of effective radiant heating surface over 60 percent of the total furnace wall area, the heat input may be increased 200 Btu/hr/cu ft of furnace volume but shall not exceed a total increase of 8,000 Btu/hr/cu ft.

3.7.4.2 Surface heat input rates. The maximum heat input rate in Btu/hr/square feet (sq ft) of effective radiant heating surface of the furnace shall not exceed the values of table III.

TABLE III. Maximum surface heat input rates.

Rated capacity, (Thousand Btu/hr)	Btu/hr/sq ft
less than 30,000	80,000
30,000 to 59,000	100,000
60,000 to 89,000	115,000
90,000 to 125,000	125,000

3.7.5 Pressure regulation. The control systems shall maintain the steam pressure at the outlet connection within ±3 percent. This tolerance shall apply to any load which, within a 1-minute period, swings from a steady-state condition to an increase (or decrease) in load equal to a maximum of 10 percent of the full rated capacity of the unit. Regulation tolerances shall apply to any steady-state condition within the burner turndown ratio (see 3.11).

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3.7.6 Exit temperature at stack. When an economizer is specified (see 3.14), the flue gas temperature leaving the economizer shall be not less than 300°F when operating above 40 percent load.

3.7.7 Noise. When specified (see 6.2.1), an assessment of the potential hazard of noise exposure onsite of the installation shall be performed by the contractor in accordance with ABMA, the Measurement of Sound from Boiler Units, Bottom Supported, Shop of Field Erected, requirements. Equipment which produce sound pressure levels of 85 decibel (dB) or greater shall be labeled with a hazardous noise warning decal 8 inch by 10.5 inch. Upon determination of the hazardous noise level of 85 dB or greater, the contractor shall notify the contracting officer or his authorized representative. The contracting officer shall furnish four decals, NSN 0105-00-212-6020, for each boiler. The contractor shall affix one decal on each side of the boiler.

3.8 Construction. Boilers shall be constructed for the specified design conditions and performance requirements in accordance with ASME Boiler and Pressure Vessel Code, Section I, Power Boilers, UL 726, UL 795, NFPA 85, and any other additional requirements specified herein. All tubes, headers, and drums shall be so assembled that the boiler and superheater, when furnished (see 3.8.7), can be drained dry.

3.8.1 Drums. Drums shall be fusion welded, stress-relieved, X-rayed, and complete with all necessary drum connections, manhole openings, with cover and yoke, nozzles, couplings, internal fittings, and other fittings as required for safe, efficient, reliable operation and for easy cleaning and maintenance. Drum internals shall be such as to ensure the requirements of 3.7.1. Steam drums shall not be less than 30 inches in diameter. Steam drums less than 12 feet long shall be furnished with at least one manhole; steam drums 12 feet long or over shall be furnished with a manhole at each end. The mud drum, or lower drum, of two-drum boilers shall not be less than 24 inches in diameter and shall be furnished with a manhole at each end. When furnished, multiple lower drums shall have handhole accesses sufficient in size and number to permit inspection and plugging of tubes and the cleaning and rolling of tubes into the drums by means of commercially available equipment. Inspection ports shall be provided in the casing to permit inspection of lower drum handhole plates while the boiler is firing. No part of a boiler drum or header surface shall be directly exposed to furnace radiation unless it is adequately protected by a refractory or a water screen in such a manner that no part of the exposed drum or header surface shall be more than 2 inches from the nearest watertube.

3.8.2 Tubes. Tubes (see 6.2.1) shall be either electric-resistance welded or seamless steel and of sufficient size to provide for free natural circulation in the proper direction for all loads. Tube bends shall be formed in a manner to permit cleaning of the internal surface of tubes with conventional tube cleaning tools. Unless otherwise specified (see 6.2.1), boiler and furnace tubes shall be not less than 2 inches in outside diameter. Membrane or fin tube or flat stud wall construction (see 6.1.3.3) shall be limited to a minimum fin thickness of 0.25 inch thick and not more than 1 inch clearance between tubes (i.e., 2-inch tubes on 3-inch centers).

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3.8.3 Refractories. Refractories shall, in accordance with ASTM C64, withstand the temperature existing under maximum load conditions. The refractories shall be formed in place or cast in sections of such size as to be easily replaceable through existing openings. The refractory sections shall be designed and secured in position in such a manner as to withstand the vibration and shocks occurring during shipment, and withstand the burner combustion pulsations.

3.8.4 Access door and observation ports. The boiler shall be equipped with a refractory lined access door for internal inspection and repairing of the furnace fireside. Observation ports shall be provided to permit operators to see the furnace interior and the entire flame pattern. Observation ports shall be fitted either with safety shields to maintain closure unless held open by the operator, or with air-cooled, heat-resistant, glass enclosures. Access to the furnace may be through the burner opening provided that the burner is of the hinged, swing-open type, having a minimum of 15-inch diameter unrestricted opening. Openings of suitable size and number shall be provided for inspection and cleaning of the entire convection tube bank. The access door and all observation ports shall be gas tight.

3.8.5 Baffles. Baffles, as required, shall be arranged to diffuse the gas properly over heating surfaces in order to obtain maximum heat absorption. The baffles shall be held securely in place without being affected by or interfering with the free expansion or contraction of the boiler. The baffles shall be constructed of refractory not less than 2-inches thick or of steel not less than 0.25 inch thick, or of heat-resistant alloy not less than 0.125 inch thick. Tangent tube baffles may be used in lieu of refractory or metal plate baffles. Baffles shall withstand the temperature under maximum load conditions.

3.8.6 Combustion chamber. The combustion chamber setting, walls, and floor shall be constructed to permit free expansion of the boiler without undue stress on any part of the boiler setting. The furnace sidewall, roof, and floor shall be water-cooled-type construction. Waterwall headers and supply tubes shall be protected from the products of combustion. Furnace floor tubes shall be covered with a minimum of 1.50 inches of refractory. All materials in the furnace area shall be of the quality and thickness used in standard commercial practice for this type of boiler. When a superheater is furnished (see 3.8.7), the length of the combustion chamber shall be sufficient to prevent flame impingement on the superheater tubes and to prevent damage to the superheater during the period between light off and steaming. For boilers having the listed rated capacities in Btu/hr, furnaces shall not exceed the shape factors in table IV. The shape factor, calculated in feet, shall equal the furnace volume divided by the product of the width and height of the furnace. Furnace volume shall be as defined in 6.1.3.1.

3.8.7 Superheater. When specified (see 6.2.1), a superheater shall be provided. The boiler shall be sequenced for control sequence I, II, or III operation (see 6.1.1b). The superheater shall be furnished as an integral part of the boiler. Superheaters shall be completely drainable with materials capable of withstanding the temperatures and pressures prevailing under maximum load conditions. Superheater pressure drop and surface proportioning shall be such as to deliver steam of the specified pressure and temperature

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TABLE IV. Shape factors.

Rated capacity, (Thousand Btu/hr)	Shape Factor	Rated capacity, (Thousand Btu/hr)	Shape Factor	Rated capacity, (Thousand Btu/hr)	Shape Factor
5,000	6.5 ft	40,000	17.5 ft	90,000	31.0 ft
7,500	7.0 ft	50,000	21.0 ft	100,000	32.5 ft
10,000	8.0 ft	60,000	24.5 ft	110,000	33.5 ft
20,000	12.0 ft	70,000	26.5 ft	120,000	34.5 ft
30,000	15.0 ft	80,000	29.0 ft	125,000	35.0 ft

under maximum continuous load steaming conditions with maximum temperature variations of 10°F after steady state steam flow conditions have been established. The superheater shall be equipped with a thermocouple well on the outlet connection, a drain, a vent, an outlet nozzle, and safety valve nozzles.

3.8.8 Insulation. All exterior surfaces of the boiler, including the floor of the boiler, but excluding doors, drumheads, handholes, manholes, vents, and appurtenances, shall be covered where practicable with not less than 2 inches of fibrous glass, mineral wool, thermal block or equivalent insulation having a heat transfer coefficient at a mean temperature of 300°F not exceeding 0.45 Btu/hr/sq ft of inch thickness per °F temperature difference. The insulation shall be so formed and secured in place as to prevent sagging or displacement during shipment and operation.

3.8.9 Outer casing. The insulation shall be covered with a sheet metal jacket made of steel having a minimum thickness of 0.1046 inch with corrugated structural steel, or structural steel vertical struts on not more than 27-inch centers, or a combination of vertical and diagonal struts which provide equivalent reinforcement. The average surface temperature of the outer casing shall not exceed 130°F with an ambient temperature of 70°F and a surface velocity of 2 ft per second, while the boiler is operating at full capacity.

3.8.10 Inner casing. The inner casing shall be pressure tight to prevent the product of combustion from coming in contact with the boiler insulation and outer casing. The inner casing may be either the boiler wall itself, made pressure tight by welding tangent tubes or finned tubes together to form a continuous surface, or a separate inner casing not less than 0.1345 inch thick placed just behind the tubes or the refractory backing, if used. The inner casing, when tested in accordance with 4.4.2.2, shall hold a pressure equal to 1.50 times the predicted maximum furnace operating pressure.

3.9 Boiler trim. Each boiler shall be fully equipped with the trim required under the applicable section of the ASME Boiler and Pressure Vessel Code plus the additional appurtenances specified herein. When disassembly is required for shipping clearance and prevention of damage during shipment, such trim and appurtenances may be field installed. The boiler trim, valves, connecting piping, and fittings shall be suitable for the design pressure and temperature as specified herein. The trim and appurtenances shall include but not necessarily be limited to the following:

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a. A pressure gage with a minimum diameter of 6 inches having a graduated range that will result in the normal pressure indication being within middle range of the scale. Pressure gages shall include a syphon, gage cock, and test connection, and shall be mounted on the boiler shell. Pressure gages shall be located at a point providing convenient reading.
b. An ASME-rated safety valve(s) as required by ASME Boiler and Pressure Vessel Code; safety valve for superheater, when applicable.
c. A water column complete with sight glass, try cock, gage cocks, and blowdown valves. The try cock and valves located more than 6 ft above floor level shall be equipped with levers, chain lengths, and handles for floor level operation. The water column shall be equipped with means to sound low- and high-water level horn alarm. Drain line piping for water column and gage glass shall terminate not more than 2 ft above operating floor with a manual valve.
d. Boiler drain valve and superheater drain valve, when applicable, sized in accordance with the ASME Code.
e. One slow-opening and one quick-opening blowoff valve or a unit tandem blowoff valve. Blowoff valves shall meet the requirements of MIL-V-18406.
f. Boilers shall be equipped for continuous blowdown with internal piping and valve with stainless steel disk and seats. The valve shall incorporate means to enable valve to be set as desired.
g. Low-water level fuel cutout bypass momentary pushbutton station. The pushbutton station is to be located adjacent to the water column and auxiliary low-water cutout drain line piping with manual valves. The pushbutton station shall provide for uninterrupted boiler operation for all fuels during either water column or low-water level fuel cutout chamber blowdown.
h. A feedwater thermometer and well with a scale equivalent to 1.5 times the saturation temperature of the steam.
i. Air vent valve with screwed connection and stainless steel disc and seats to vent entrapped air from the boiler and air vent valve for superheater, when applicable.
j. Main steam nonreturn valve of the angle or Y pattern, stop check type with flanged connections. Valve shall be sized on the basis of the maximum steam generating capacity and shall not be of a size less than the boiler steam outlet size.
k. A chemical feed connection with connected internal distributing pipe.
l. A steam main with a calorimeter connection.
m. When specified (see 6.2.1), connection for remote sensing of steam pressure, steam flow, steam temperature, air flow, water and/or fuel temperatures.
n. When specified (see 6.2.1), a recorder shall be furnished to indicate and record conditions, such as steam flow, air flow, steam pressure, and steam and water temperatures as specified (see 6.2.1). Unless otherwise specified (see 6.2.1), the recorders shall be mounted on the control panel.
o. Two low-water level cutout switches of the float- or electrode-actuated type may be integral with or separate from the water column (see 3.9.1), and require a manual reset and shall activate a red indicating light.

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p. Excessive steam pressure limit switch to be supplemental to and separate from the operating control. The limit switch may recycle on control sequence I and shall cause a nonrecycling safety shutdown on control sequences II and III and require a manual reset and activate a red indicating light.

3.9.1 Feedwater regulator. Each boiler shall be furnished with a feedwater regulator conforming to the requirements of MIL-R-18115. Unless otherwise specified (see 6.2.1), regulators shall be type I, class I, style 1. When specified for boilers servicing unusual service load characteristics or power generation service (see 6.2.1), two- or three-element regulators shall be provided.

3.10 Control systems. Boiler control systems shall conform to MIL-B-18796 and shall at least meet the applicable requirements of NFPA 85. Boilers shall be furnished with the following control systems:

a. Programming control system to provide the required control sequence specified in 1.2 in accordance with MIL-B-18796. The programming control system shall include repetitive self-checking circuits.
b. Safety control system including flame safeguard control, boiler limit controls, and fuel safety interlocks specified herein and in MIL-B-18796.
c. Modulating combustion control system shall be as specified in MIL-B-18796.

Unless otherwise specified (see 6.2.1), the combustion control system may be electrically, pneumatically, or hydraulically operated.

3.11 Burners and fuel systems. Boilers shall be furnished with factory installed burners, ignition systems, and associated fuel systems in accordance with the requirements and options of MIL-B-18796. The burners and fuel systems shall be designed for the fuel systems specified in 1.2. The supply temperature shall be as specified (see 6.2.1). Unless otherwise specified (see 6.2.1), minimum burner turndown shall be 5 to 1.

3.11.1 Oil burners. Oil-fired units shall have oil burners of the types specified in MIL-B-18796.

3.11.2 Gas-fired burners. Gas-fired units shall have power burners of the partial premix or of the postmix type conforming to MIL-B-18796.

3.11.3 Combination gas- and oil-fired units. A combination burner shall consist of an oil burner and a gas burner each conforming to the requirements of MIL-B-18796 separate oil and gas burners.

3.11.4 Type GLOHO units. Type GLOHO units shall have burners conforming to MIL-B-18796, class 6.

3.11.5 Fuel oil pump and strainers. Unless otherwise specified (see 6.2.1), types LO, HO, GLO, GHO, and GLOHO boilers shall be furnished with a fuel oil pump and strainers conforming to the requirements of MIL-B-18796.

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3.11.6 Fuel oil heaters. Unless otherwise specified (see 6.2.1), types HO, GLOHO and GHO boilers shall be furnished with fuel oil heaters conforming to the requirements of MIL-B-18796.

3.12 Draft fans. The draft fans shall conform to the draft fans of MIL-B-18796. The draft fan shall be for a pressurized or balanced draft system, as specified (see 6.2.1).

3.13 Control panel. The controls, including operating switches, indicating lights, gages, alarms, motor starters, fuses, and circuit elements of control systems shall be mounted on a single control panel or cabinet insofar as practicable in order to centralize the control functions conforming to MIL-B-18796. Burner and boiler controls shall be located in the control panel. The control panel may be either boiler front mounted or freestanding (see 6.2.1). When a freestanding control panel is to be furnished, construction details covering cabinet and wiring for remote starting location, mounted controls and instruments shall be as specified (see 6.2.1).

3.13.1 Annunciator. When specified (see 6.2.1), an annunciator shall be provided for the boiler in accordance with MIL-B-18796. Unless otherwise specified (see 6.2.1), the annunciator may either be a separate item, or an integral part of the control panel or burner control system.

3.13.2 Smoke density indicator and transmitter. An electronic opacity monitoring device with strip chart recorder and transmitter for stack mounting shall be provided on type HO, LO, GLO, GHO, and GLOHO to detect, indicate, alarm, and record visible emissions from the stack in accordance with MIL-B-18796.

3.14 Economizer. When specified (see 6.2.1), a separately packaged economizer shall be furnished. The economizer may be equipped with plain or finned tubes and shall be complete with prefabricated interconnecting breeching for attachment to the boiler flue outlet and feed piping between the economizer and the boiler drum shall require minimum field assembly. Finned tube economizers, when supplied with type HO or GHO boilers, shall not have more than three fins per inch of tube length. Valving shall be provided to allow bypass of the feedwater around the economizer. The economizer shall be designed and constructed in accordance with ASME Boiler and Pressure Vessel Code, Section I. Materials used in construction shall withstand the temperatures and pressures prevailing under maximum load conditions.

3.15 Soot blowers. When type HO or GHO boilers are required (see 1.2 and 6.2.1), soot blowers shall be furnished with boilers, and with economizers and superheaters if required in 3.8.7. Soot blowers furnished with boilers and economizers shall be steam operated, valve-in-head type and shall be furnished complete with wall sleeves, clamps, hangers, supports, operating chains, and other appurtenances required for a complete installation. The blower elements shall be so arranged that all parts of the heating surfaces shall be cleaned of any soot deposits when rotated by manual or automatic means. Elements shall be of such length, diameter, and total nozzle area that for the operating pressure involved there will be no significant difference in the cleaning effect between the nozzle nearest the inlet and those farthest from the inlet of the element. Soot blower piping shall include a valve chain

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operated from the floor, in addition to an auxiliary steam-stop valve, and drain connection. Soot blowers shall be fitted with scavenging air ports and shall be connected by tubing to the draft fan in a manner to provide sufficient air for continuous scavenging of the blowers while the boiler is firing during periods when soot blowers are not operated for cleaning purposes. Soot blowers furnished with superheaters shall be steam operated and retractable. Soot blower bearings and wall boxes shall be provided on type LO, G, and GLO boilers to permit future installation of soot blowers.

3.16 Stack. When specified (see 6.2.1), stack sections, stack supports, umbrella collar and cap, and flue transition piece (if required in 3.17), shall be provided. Stack height shall be as specified (see 6.2.1).

3.17 Transition piece. When specified (see 6.2.1), a transition piece or junction piece shall be furnished to permit adapting from the boiler outlet to stacks or breeching. Transition pieces shall be designed for vertical or horizontal discharge as specified (see 6.2.1). The transition piece for vertical discharge shall be capable of carrying the full weight of the stack.

3.18 Electrical wiring and equipment.

3.18.1 Electric motors. Electric motors for the driven auxiliaries shall be constructed and rated in accordance with the applicable provisions of NEMA MG 1. Motors shall be general purpose, continuous duty, ball bearing, induction motors. The motors shall be designed for operation on a power supply having the specified characteristics (see 6.2.1). Unless otherwise specified (see 6.2.1), motors shall be furnished with dripproof enclosures and encapsulated or sealed windings.

3.18.2 Motor starters. Unless otherwise specified (see 6.2.1), motors shall be provided with magnetic across-the-line starters with overload and undervoltage protection and manual-off-automatic control stations. Motors over 15 hp shall have reduced voltage starting with overload and undervoltage protection and manual-off-automatic control stations. Magnetic starters shall conform to the applicable requirements of NEMA ICS 1, ICS 2, and ICS 6, and shall have NEMA-type 12 enclosures.

3.18.3 Wiring and safety devices. All electrical wiring and circuit safety devices shall conform to NFPA 70. All control circuits shall be 2-wire, 1-side-grounded, not to exceed nominal 120 volts, and be protected with suitable fuses, or circuit breakers.

3.19 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

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3.20 Electromagnetic interference suppression. When specified (see 6.2.1), the unit shall conform to the electromagnetic interference suppression requirements and test limits for class C3, group III equipment as specified in MIL-STD-461.

3.21 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2.1). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.21.1 Aluminum heat-resisting painting. When specified (see 6.2.1), before assembly, the boiler shell and the interior of the jacket shall be finished with one coat of aluminum heat-resisting paint conforming to DOD-P-24555, with manufacturer's certification that the material will perform satisfactorily when air-dried at ambient tempeature and subsequently subjected to 100°F continuous exposure in use. The paint shall be applied directly to a commercial-blasted steel surface conforming to SSPC-SP6, to a minimum dry film thickness of 1.5 mils. The surface shall be painted the same day as cleaned. After assembly, any damaged areas shall be recleaned and spot-primed before applying the second 1.5-mil dry-film-thickness coat.

3.22 Lifting and tiedown attachments. The boiler shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the boiler. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the boiler on the carrier when shipped.

3.23 Commercial technical manuals. When specified in the contract or order, commercial technical manuals shall be prepared (see 6.2.2).

3.24 Operating instructions. When specified in the contract or order, operating instructions shall be prepared in a permanent, legible form suitable for posting on or near the boiler (see 6.2.2). The operating instructions shall include information relating to the normal ignition and shutdown of the unit and shall otherwise comply with the recommendations for operating instructions as given in UL 726, or UL 795, or both, as applicable.

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3.25 Identification marking. Identification shall be permanently and legibly marked directly on the boiler or on a corrosion-resisting metal plate securely attached to the boiler at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name, and trademark to be readily identifiable to the manufacturer.

3.25.1 Identification plate. An identification plate will be furnished by the contracting officer for each boiler. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each boiler in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.26 Workmanship and fabrication. Workmanship, welding, and fabrication shall be in accordance with the ASME Boiler and Pressure Vessel Code and shall be of such quality as to produce boilers that meet the standards prevalent among manufacturers who normally produce equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsiblity for compliance. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).
c. On-site inspection (see 4.2.3)

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4.2.1 First article inspection. When specified (see 6.2.1), a first article inspection shall be performed on one boiler when a first article inspection is required (see 3.2 and 6.3). This inspection shall include the examination of 4.3 the tests of 4.4.1 through 4.4.1.3, the electromagnetic interference suppression test of 4.4.5, the hydrostatic tests of 4.4.2, and when specified (see 4.6 and 6.2.1) the preproduction pack inspection of 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the on-site inspection of 4.2.3, the design verification of 4.5 and the packaging inspection of 4.6.

4.2.3 On-site inspection. Testing shall be performed on all boilers at the site after installation and shall be witnessed by a certified boiler inspector acceptable to the contracting officer. This testing shall be in addition to quality conformance inspection performed at the factory. On-site testing shall consist of the tests of 4.4.3, the factory auxiliary equipment tests of 4.4.4, and all other tests deemed necessary by the procuring activity when specified in the contract to verify compliance with the requirements of this specification. On-site testing shall be performed either by the boiler manufacturer or by the installing contractor, as specified (see 6.2.1). The manufacturer shall have the privilege of representation at tests performed by others. When the manufacturer is responsible for on-site testing, the detailed requirements and schedule for the test program will be as specified in the contract (see 6.2.1). In all cases, deficiencies revealed by on-site testing shall be corrected at the boiler contractor's expense and any required retesting shall also be at the supplier's expense.

4.3 Examination. Each boiler shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements preventing or lessening maximum efficiency or any defect shall constitute cause for rejection.

4.4 Tests.

4.4.1 Fire and explosion safety tests. Tests shall be conducted as necessary to determine compliance with the applicable UL Safety Standards and system safety requirements herein. The presence of the applicable UL or, when specified in the contract or order, a certificate of compliance (see 6.2.2) required for MIL-B-18796 will be accepted as evidence of compliance.

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4.4.1.1 Oil-fired boilers. Oil-fired boilers with capacity below 12.5 million Btu/hr shall meet the test requirements of UL 726. Oil-fired boilers with capacity of 12.5 million Btu/hr and above shall meet the test requirements of NFPA 85. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2) and will be accepted as evidence of compliance.

4.4.1.2 Oil burners. Oil burners with capacity below 12.5 million Btu/hr shall meet the test requirements of UL 296. Oil burners with capacity of 12.5 million Btu/hr and above shall meet the test requirements of NFPA 85. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2) and will be accepted as evidence of compliance.

4.4.1.3 Gas-fired boilers and gas burners. Gas-fired boilers and gas burners with capacity below 12.5 million Btu/hr shall meet the test requirements of UL 795. Gas-fired boilers and gas burners with capacity of 12.5 million Btu/hr and above shall meet the test requirements of NFPA 85. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2) and will be accepted as evidence of compliance.

4.4.2 Hydrostatic tests.

4.4.2.1 Watersides including fitting and accessories. Watersides shall be hydrostatically tested in accordance with the requirements of the ASME Boiler and Pressure Vessel Code. The ASME stamp will be accepted as evidence of this test.

4.4.2.2 Boiler casing, air casings, and ducts. Air casing and ducts exterior to the furnace, shall be tested pneumatically at the maximum working pressure. The soap bubble or the halogen-sniff test method shall be used to verify tightness. Combustion chambers and flue-gas passages of boilers normally operated under pressure shall be tested for tightness at 1.5 times the predicted operating pressure in the furnace at maximum continuous output. For this test, the boiler shall be tightly sealed with a suitable means to blank off all openings. Air shall be admitted to the boiler until the test pressure is reached. The outer casing shall be vented to the atmosphere during the casing pressure test. The air pressure shall be held in the boiler during the test. If in a 10-minute period the pressure drop does not exceed 1 inch water gage, the casing shall be regarded as tight and accepted. Failure to pass the requirements of 3.8 and this test shall constitute cause for rejection.

4.4.2.3 Fuel systems. The preassembled fuel system that is furnished integrally with the boiler shall be checked for leakage during the fire testing of the boiler. The soap bubble or the halogen-sniff test method shall be used to verify integrity of the gas fuel system. Failure to pass the requirements of 3.11 and this test shall constitute cause for rejection.

4.4.3 On-site inspection tests.

4.4.3.1 Capacity and efficiency tests. The capacity, efficiency, and steam quality at the specified capacity of the boiler shall be determined in accordance with the ASME Performance Test Code PTC 4.1. The ASME Test Form for Abbreviated Efficiency test may be used. The efficiency shall be

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determined by the heat loss method. Performance shall be as specified in 3.7. Methods of measuring the flow of steam, water, and fuel other than weighing will be allowed. The accuracy of the method shall assure the Government the desired performance specified in 3.7 and be acceptable to the contracting officer or his authorized representatives. Test runs shall be made at the maximum capacity.

4.4.3.2 On-site operational tests. The boiler shall be tested to demonstrate control and operational conformance to the requirements of this specification and MIL-B-18796, under varying load conditions ranging from the specified capacity to the minimum burner turndown ratio without on-off cycling. All burners shall be tested for ability to start a cold boiler in accordance with the manufacturer's instructions. Unless otherwise specified (see 6.2.1), oil burners shall be tested for ability to make a cold start from the specified oil supply temperature of 3.11. Automatic burners shall also be tested for ability to provide steam in accordance with steam demand when on-off cycling is required. Boilers that exhibit (1) excessive or unexplained loss of ignition, (2) nuisance shutdown due to faulty burner or control operation, (3) improper flame, (4) excessive carbon deposits, or (5) necessity for frequent or difficult adjustments shall be rejected. Operational tests shall include the following:

4.4.3.2.1 Sequencing. The boiler shall start, operate, and stop in strict accordance with the specified operating sequence.

4.4.3.2.2 Burner tests. Tests shall be conducted on the burner as specified in section 4 of MIL-B-18796.

4.4.3.2.3 Boiler limit and cutout switches, and fuel safety interlocks. Safety shutdown shall be caused by simulating interlock actuating conditions for each boiler limit and cutout switch, and fuel safety interlock. Safety shutdowns shall occur in the specified manner.

4.4.3.2.4 Safety valves. The high pressure limit switch shall be locked out or otherwise made inoperative and the boiler safety valves shall be lifted by steam. The relieving capacity, popping pressure, blowdown, and reseating pressure shall be determined by observation and measurement to be in accordance with the ASME Boiler and Pressure Vessel Code. This test shall be made in the presence of the contracting officer or his authorized representative. The ASME standard symbol will be accepted only as indicating compliance with the design and material requirements of the code.

4.4.3.2.5 Blowdown valves and try cocks. All blowdown valves and try cocks shall be tested for proper operation.

4.4.3.2.6 Tests for particulate and nitrogen oxide emissions. When specified (see 6.2.1), the tests for emissions of particulates, nitrogen oxide, and sulfur dioxide shall be conducted in accordance with MIL-B-18796.

4.4.4 Factory auxiliary equipment tests. Draft fans, fuel oil heaters, fuel pumps, and electric motors shall be tested when necessary to determine compliance with the referenced standards. Standard symbols and certifications (see 6.3) from the referenced organization may be accepted at the discretion

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of the contracting officer or his authorized representative. The operation of fans, fuel oil heaters, fuel pumps, and electric motors shall be closely observed during onsite operational tests (see 4.4.3.2) for possible defects or nonconformance. The action of dampers, actuators, and valves shall be without backlash.

4.4.5 Electromagnetic interference suppression tests. To determine conformance to 3.20, the first unit of production shall be subjected to the tests in accordance with MIL-STD-462. When suppressed to conform to 3.20, the manufacturer, upon approval of the contracting officer and when specified in the contract or order (see 6.2.2), shall prepare and furnish a certificate of compliance, in lieu of the test, that the boiler meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.5 Design verification. When specified in the contract or order (see 6.2.1 and 6.2.2), the manufacturer shall prepare drawings and data necessary to verify compliance with the heat release criteria of 3.7.4, 3.7.4.1, and 3.7.4.2.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-B-3180. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2.1), a preproduction pack shall be furnished for examination and test within the timeframe required (see 6.2.1).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-B-3180 with the level of preservation and the level of packing as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The boilers covered by this specification are intended for generation of steam to be used in processing operations, space heating, or power generating systems on land installations.

6.1.1 Selection factors. The following should be considered in the selection of a boiler and the exercise of options herein:

a. The boiler type (oil, gas, or combination) should be determined in accordance with the availability and economics, the fuel supply, and any DOD requirements for capability of firing multiple fuels.
b. Automatic control sequences should not be specified for boilers furnished with superheaters. Superheaters should be limited to boilers with means for starting with low fire until saturated steam enters the superheater and then manually raising the combustion rate to high fire to prevent overheating. Superheaters can be provided to furnish steam up to 300°F above saturation temperatures.
c. Repetitive self-checking circuits have been included as standard because operating in occupied building or in hazardous locations require such safety controls.

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d. Normally, heavy oil-fired boilers above 5,000,000 Btu/hr output should be capable of burning fuel oil grades No. 2, No. 4, No. 5, and No. 6. without difficulty. The pump and motor drive selected for each burner should be sized based on firing the heaviest grade of oil because of the higher viscosity. When No. 5 and No. 6 oil is to be fired the burner should be provided with a preheater to reduce the oil viscosity. When it is required to fire more than one grade of oil, as in class 6 burners, separate fuel storage systems and fuel trains are required. Care should be exercised when more than one oil is to be used due to possible incompatibilities from different bases (i.e., paraffin and asphalt). The fuel train for firing No. 6 oil requires a pump and a heat exchanger (in addition to the preheater at the burner) to heat the oil to the point it can be pumped. The fuel train shall incorporate a separate return line to the storage tank. Where temperatures drop below 40°F a suction stub heater installed in the storage tank will be required with heat traced oil lines to keep the No. 6 oil in the pumpable range.
e. Gas-electric igniters as specified herein are required for UL, FM, and Industrial Risk Insurers requirements. Oil igniters should be specified only when gas is not available.
f. If the boiler is subject to environmental regulations, each agency should consult with their environmental support office since regulation limits of particulate and emissions vary from city, state, and Federal governments.
g. Economizers can increase fuel to steam efficiencies from 2.50 to 3.00 percent and will be required on higher pressure boilers to achieve desired efficiency.
h. Economizer shall be supplied with feedwater at a minimum temperature of 240°F to preclude condensation of flue gases.
i. Typically boilers up to 40,000 Btu/hr are are furnished with boiler front mounted control panels.

6.1.2 Other fuels. The performance requirements specified herein may be based on other fuels providing the quantity on site is sufficient for tests specified in section 4 and a guaranteed ultimate analysis, higher heat content, and oil viscosity are made available at time of testing the boilers. Corrections between the values of the analyses in section 3 of this specification and the guaranteed analyses and heating value should be made in accordance with section 7 of ASME PTC 4.1. Referenced below are some of the most commonly used fuels with their viscosity range and average higher heat content:

Fuel	Viscosity range at 100°F	Average higher heat content
Light oil Grade 2 ASTM D 396 JP-5 MIL-T-5694 Diesel VV-F-800	32-38 SSU	136,000 Btu/gal

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Fuel	Viscosity range at 100°F	Average higher heat content
Heavy oil		
Grade 4 ASTM D 396	45-125 SSU	150,000 Btu/gal
Grade 5 ASTM D 396	150-750 SSU	150,000 Btu/gal
Navy Special MIL-F-859	150-400 SSU	150,000 Btu/gal
Grade 6 ASTM D 396	1,000 SSU	150,000 Btu/gal
	Specific gravity	
Natural gas	0.65	1,076 Btu/cu ft
Manufactured gases	0.38	535 Btu/cu ft
Mixed gas	0.50	800 Btu/cu ft
Liquified petroleum gas	1.55	2,500 Btu/cu ft

6.1.3 Definitions.

6.1.3.1 Furnace volume. Furnace volume shall be defined as the cubic contents of the space provided for combustion of the fuel. Furnace limits shall be defined as limits not extending beyond the first plane of entry into, or between, the tubes. Screen tubes, when used, shall be considered as the first plane of entry. The space occupied by a radiant type superheater (if used) shall not be considered furnace volume.

6.1.3.2 Effective radiant heating surface. Effective radiant heating surface is defined as heat exchange surface, exclusive of superheater elements, which is directly exposed to the radiant heat of the flame on one side and to the water or water-steam mixture being heated on the other. Effective radiant heating surface shall be calculated on the side receiving heat and shall consist of plain or finned tubes. Except for 50 percent of refractory-covered floor tubes, refractory-covered surfaces shall not be counted in the computation of the effective radiant heating surface. Computations shall be based on the following considerations:

a. The flat projected area of bare tubes shall be considered as effective radiant heating surface.
b. Metallic surfaces extending from tubes or headers shall be considered to have an effective radiant heating surface equal to 90 percent of their flat projected area except that the following extended surfaces shall not be considered as effective radiant heating surface:

(1) Extended surfaces less than 0.25-inch thick.
(2) Extended surfaces longer than 1 inch.

c. The flat plane area of the furnace exit plane shall be considered as effective radiant heating surface.

6.1.3.3 Membrane wall construction. A furnace closure wall made up of closely spaced waterwall tubes welded together or to an intermediate fin to form a continuous airtight structure.

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6.1.3.4 Tangent tube wall construction. A waterwall in which the tubes are substantially tangent to each other with practically no space between the tubes. Also refered to as "tube-to-tube wall."

6.1.3.5 Fin tube wall construction. Spaced waterwall tubes on which continuous flat metal extensions are welded in a plane parallel to the wall.

6.1.3.6 Stud-tube wall construction. A wall containing watertubes covered with refractory which is held in place by stud anchors attached to the tubes.

6.1.3.7 Flat stud wall construction. Spaced waterwall tubes on which separated flat metal extensions are welded in a plane parallel to the wall.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, date, and number of this specification.
b. Firing system type with fuel type, and control sequence (see 1.2.1 and 1.2.2).
c. When D-type boilers shall be either right- or left-hand (see 3.1).
d. When a first article inspection is required (see 3.2, 4.2.1 and 6.3).
e. Design conditions and capacity (see 3.6, and 3.7.2).

    (1) Maximum continuous steam flow, lb/hr (see 3.6a).
    (2) Boiler design pressure (see 3.6b).
    (3) Operating pressure (psig) at steam drum or superheater outlet, as applicable (see 3.6c).
    (4) Steam temperature (°F) (see 3.6d).
    (5) Feedwater temperature (°F) (see 3.6e).
    (6) Boiler site elevation (feet) (see 3.6f).
    (7) Maximum and minimum anticipated ambient air temperatures (°F) (see 3.6g).
    (8) Maximum continuous output (Btu/hr) (see 3.6h).

f. When performance criteria shall be based on other than as specified. Supply the required fuel characteristics (ultimate analysis, and higher heating value) of intended oil or gas fuel (see 3.7).
g. Available gas supply pressure (see 3.7) (should be obtained from the gas utility company).
h. Boiler capacity required (see 3.7.2).
i. When the boiler shall conform to particulate, nitrogen oxide, and sulfur dioxide emissions requirements, and emission limits (see 3.7.3.1 and 6.1.1f).
j. When an assessment of the potential hazard of noise exposure is required (see 3.7.7).
k. When tubes shall be of a size other than as specified (see 3.8.2).
l. When superheaters are required (see 3.8.7).
m. Required connections for remote sensing of steam pressure, steam flow, air flow, water temperatures, and fuel(s) (see 3.9m).

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n. When boiler conditions are to be recorded, if recorders are to be furnished, and if recorders are to be mounted on other than the control panel (see 3.9n).
o. When feedwater regulator shall be other than as specified (see 3.9.1).
p. Medium of operation (control systems) for combustion control if different (see 3.10).
q. The required fuel supply temperature (see 3.11).
r. When fuel pump is different (see 3.11.5).
s. When fuel oil heaters are different; when duplex pump and heater sets are required (see 3.11.6).
t. Type of draft system required (see 3.12).
u. Type of control panel required: Required location of control panel and details of the freestanding control panel including location, construction, mounted controls, instruments, and wiring for remote starting (see 3.13).
v. When an annunciator is required and if different (see 3.13.1).
w. When an economizer is required (see 3.14).
x. When soot blowers are required (see 3.15).
y. When a stack is required and height (see 3.16).
z. When a transition piece is required and whether for vertical or horizontal discharge (see 3.17).
aa. Required electrical characteristics of the motor and if motor enclosures are different (see 3.18.1).
bb. If motor starters are to be other than as specified (see 3.18.2).
cc. When fungus resistance is required (see 3.19).
dd. When electromagnetic interference suppression is required (see 3.20).
ee. When painting is to be other than manufacturer's standard; color of finish coat required (see 3.21).
ff. When surfaces are to be painted with aluminum heat-resisting paint conforming to TT-P-28 (see 3.21.1).
gg. When test requirements of 4.2.1 shall be different from the onsite inspection (see 4.2.3).
hh. Whether onsite inspection will be performed by the boiler manufacturer, or by the installing contractor; outline of tests if boiler manufacturer will perform the tests (see 4.2.3).
ii. When oil burners shall not be tested for their cold start ability at oil supply temperature (see 4.4.3.2).
jj. When boiler shall be tested for emissions of particulates and oxides of sulfur and nitrogen; method of testing; who is to perform tests (see 4.4.3.2.6).
kk. When design verification is required (see 4.5).
ll. When a preproduction pack inspection is required and timeframe required for submission (see 4.6).
mm. Level of preservation and level of packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved

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Contract Data Requirements List (CDRL) incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following papagraphs.

Paragraph No.	Data requirements title	Applicable DID No.	Option
3.23, 3.24	Commercial manual	DI-M-4022	------
4.4.1 4.4.1.1 4.4.1.2 4.4.1.3	Certificate of compliance	DI-E-2121	------
4.4.5	Certification data/ report	UDI-A-23264	------
4.5	Drawings, Engineering and Associated Lists	DI-E-7031	Level 3 Design activity designation Contractor Drawing No. Contractor
4.5	Design data and calculation	DI-MISC-80296	------

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross-reference. The following cross-reference of classification applies.

MIL-B-17095G	MIL-B-17095F
Type GLOHO	None
Control sequence III	Control sequence IV
Deleted	Boiler groups 1-5

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6.5 Subject term (key word) listing.

Boiler
Combination gas and oil fired
Gas fired
Oil fired
Output - 5,000,000 to 125,000,000 BTU/Hr
Packaged type
Steam generating
Watertube

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review Activities:
Navy - SA
Army - CE
Air Force - 84

Preparing Activity:
Navy - YD

(Project 4410-0059)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

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The contract or order (see D.6.2) shall provide any [illegible]

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-17095G	BOILERS, STEAM, WATERTUBE, PACKAGED TYPE (OUTPUT 5,000,000 TO 125,000,000 BTU/HR, OPERATING PRESSURE 100 TO 900 PSIG)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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